SoK: Decentralized Finance (DeFi) - Fundamentals, Taxonomy and Risks

KRZYSZTOF GOGOL, Department of Informatics IfI, University of Zurich UZH, Switzerland
CHRISTIAN KILLER, Department of Informatics IfI, University of Zurich UZH, Switzerland
MALTE SCHLOSSER, Department of Banking and Finance DBF, University of Zurich UZH, Switzerland
THOMAS BOCEK, Axelra, Switzerland
BURKHARD STILLER, Department of Informatics IfI, University of Zurich UZH, Switzerland

Financial services that are not only related to crypto-currencies but rely on blockchain for security and integrity are jointly referred to as Decentralized Fiance (DeFi) and are evolving rapidly. Given their novel applications of DLT and sophistical economical designs, the distinction between DeFi services and understanding the involved risk are often complex.

This paper systematically studies the major classes of DeFi protocols, including risk and security. The selection of DeFi categories is based on a quantitative approach, covering over 80% of total value locked (TVL) in DeFi. Further, a structured methodology is provided to differentiate between DeFi protocols based on the algorithmic design and blockchain-network architecture. The findings indicate that every DeFi protocol falls into one of the three classes of DeFi algorithms: liquidity pool, synthetic asset or aggregator protocol. This work concludes with the risk analysis that is derived from the DeFi protocol, underlying tokens and agents. Certain DeFi assets, such as crypto-backed stablecoins, liquid staking tokens, and wrapped tokens of bridges, are synthetic assets, similar to derivatives in traditional finance, and bore similar risk exposure.

Additional Key Words and Phrases: Decentralized Finance, DeFi, Blockchain, SoK

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1 INTRODUCTION

In 2008 the Global Financial Crisis put the financial markets in turmoil. In the same year, a pseudonym *Satoshi Nakamoto*, an anonymous author or a group of authors, proposed a new decentralized payments system and called it - Bitcoin[114]. Unlike the Traditional Financial system (TradFi), Bitcoin does not depend on any centralized and legal entity, such as a bank acting as a financial intermediary. Sparked by Bitcoin, a myriad of new Blockchain (BC) projects and cryptocurrencies emerged, and the total market cap of Bitcoin reached over USD 1 trillion in 2021 [23]. With the launch of the Ethereum BC [65] in 2015, a new generation of BCs emerged. In contrast to Bitcoin and its forks, those BCs allowed for the execution of turing complete Smart Contacts (SC), executed within the virtual machine hosted by the

Authors' addresses: Krzysztof Gogol, gogol@ifi.uzh.ch, Department of Informatics IfI, University of Zurich UZH, Binzmühlestrasse 14, Zürich, Switzerland, CH—8050; Christian Killer, killer@ifi.uzh.ch, Department of Informatics IfI, University of Zurich UZH, Binzmühlestrasse 14, Zürich, Switzerland, CH—8050; Malte Schlosser, malte.schlosser@bf.uzh.ch, Department of Banking and Finance DBF, University of Zurich UZH, Plattenstrasse 32, Zürich, Switzerland, CH—8032 Zürich,; Thomas Bocek, thomas.bocek@axelra.com, Axelra, Hagenholzstrasse 83, Zürich, Switzerland, CH—8050; Burkhard Stiller, stiller@ifi.uzh.ch, Department of Informatics IfI, University of Zurich UZH, Binzmühlestrasse 14, Zürich, Switzerland, CH—8050.

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decentralized BC network. To further advance decentralization, the concept of Decentralized Exchange (DEX) with Automated Market Maker (AMM) emerged in 2016 [66]. Unlike traditional stock exchanges, the AMM-based DEX does not rely on a centralized order book. This idea was successfully implemented in 2017 by Uniswap[52], the first DEX. MakerDAO[108] and Compound[121] followed with the first decentralized stablecoin and decentralized lending protocol, respectively. In 2016 perpetuals were introduced by BitMEX[53], however, still relying on intermediaries. Perpetuals resemble traditional futures contracts without a fixed expiration date to enable derivatives amrkets for illiquid assets. Based on AMMs and collateralized vaults, perpetuals without relying on intermediaries emerged in mid 2020 with MCDEX[37], the Perpetual Protocol[41], or GMX[33].

DeFi, as enabled by public and permissionless distributed ledger, differs from TradFi in various dimensions. First, *(i)* the *lack of intermediaries* allow parties in DeFi protocols to interact with each other instantly via SCs deployed on BCs. Consequently, the transaction settlement occurs within seconds, 24/7, contrary to days in TradeFi, and is more cost-efficient. Second, *(ii)* the *permissionless* nature means that any counter-party can interact with the SC, deploy its own SC, and all transactions and SCs are publicly verifiable. For example, any BC reserves or treasuries of DeFi protocols can be instantly audited. Third *(iii)*, the *immutability* of the SCs and the BC imply that *code is law* and that nothing can be changed or altered. It should be noted that certain protocols have upgradable SCs, where a certain amount of people are required to perform updates, or Decentralized Autonomous Organizations (DAOs) are voting on specific adaptions. Consequently, DeFi increases efficiency, transparency and accessibility of the financial services[125]. Furthermore, the source code of DeFi protocols is often open-source and the governance is executed via DAOs, allowing arbitrary stakeholders to participate in the decision-making process[94].

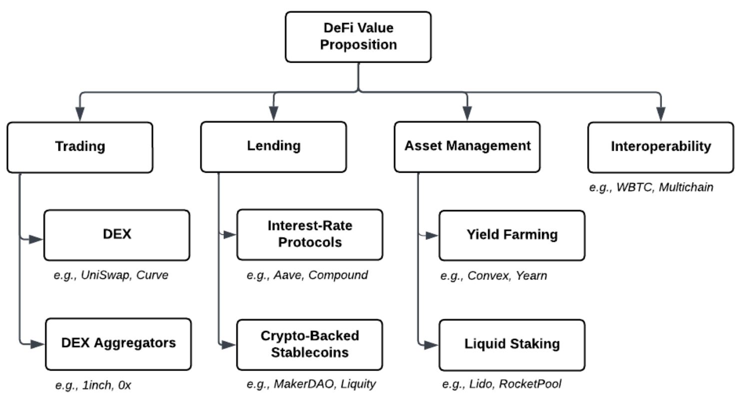

Fig. 1. DeFi Value Proposition

With general-purpose and programmable BCs (*e.g.*, Ethereum[65], Solana[21], BSC[8]), it became possible to provide advanced financial services and products without relying on intermediaries for operations, settlement or execution. DeFi applications, called DeFi protocols, allow swapping, lending, margin trading, and borrowing tokens directly within the BC, as depicted in figure 1. Tokens, referred to as digital assets, can cryptographically prove ownership of the real-world asset (*e.g.*, art, real estate, and financial products), be linked to other tokens or exist purely digitally for governance, utility, social or other purposes. Another popular process is yield farming. It describes the liquidity provisions to DeFi protocols to enable its operations in exchange for participation in the charged fees and thus serves a similar role as market-making in TradFi. The Total Value Locked (TVL), a value representing the value of digital assets locked in yield farming or other DeFi protocols as collateral, grew from zero to USD 200 billion in just two years[11] and over five million BC wallet addresses interacted with DeFi protocols[24].

The rapid adoption of the DeFi protocols quickly revealed shortcomings of both protocols and underlying BCs. High transaction costs were the main factor hindering the winder DeFi adoption on the Ethereum BC. Still, Ethereum comprises today 58.5% of total TVL in DeFi, followed by Tron[47] (11.0%) and BSC[8] (10.6%) [13]. Layer 2 BCs on top of Ethereum, *e.g.*, Polygon[42], Optimisms[39], zkSynz[25], starknet[22], emerged as scaling solutions that leverage the security of Ethereum and optimize the transaction costs by taking the calculation off-chain.

1.1 Motivation and Contribution

Numerous SoK and surveys on DeFi and BC exist, however, this work offers novel perspectives. First, related works mostly provide a high-level overview of DeFi and, second, these works arbitrarily mention selected categories of DeFi protocols[58, 68, 85, 95, 125, 132, 140]. Another group of research papers focuses on the vertical analysis of specific DeFi protocols, *e.g.*, decentralized exchanges with automated market making[134], yield farming[76, 133] or stablecoins[64, 113]. Thus, this work presents the first Systematization of knowledge (SoK) that *(i)* applies a data-driven approach for the selection of DeFi protocols' categories, and *(ii)* combines the vertical and horizontal research, in order to present a comprehensive overview of DeFi.

This work analyses the major, in terms of TVL, classes of DeFi protocols. The initial classification and TVL measures are derived from DeFi Llama[12] database. Only DeFi categories, which have TVL above five billion US dollars, are included in the analysis. Those categories represent over 89.09% of TVL in DeFi as of 18.12.2022 and are listed in the table 1. For the purpose of completeness, we also consider DEX aggregators and algorithmic stablecoins. TVL measure does not apply to DEX aggregators and algorithmic stablecoins. However, DEX aggregators initiate 25% daily traded volume via DEXs[12]. When discussing crypto-backed stablecoins, we introduce and classify all other stablecoins - fiat-backed and algorithmic. Perpetual protocols, with two billion US dollars TVL, are the largest DeFi category that does not meet the indicated treshold.

Table 1. Categories of DeFi protocols[12] with Total Value Locked (TVL) denominated in USD

DeFi Protocol Category	Protocols	TVL
DEX	648	15.86b
Lending, Interest-rate Protocols	196	10.83b
Liquid Staking	60	8.63b
CDP, Crypto-backed Stablecoins	60	7.92b
Bridge	36	7.81b
Yield, yield farming	364	5.3b
Other	656	7.18b

2 FUNDAMENTALS

The general design principles for blockchains and Distributed Ledger Technology (DLT) and their implications for activities in financial services have been widely studied [95]. This section introduces the necessary blockchain definitions that are later used in the paper.

Table 2. Terminology used in this work

Abbreviation	Term	Definition
BC	Blockchain	Public data storage executing a permissionless consensus protocol in a decentralized deployment [**?**]
SC	Smart Contract	Digital protocol implemented in software, executing the terms of a contract agreed between two or more parties without the need of an intermediary [**?**]
L1	Layer 1	BC with its own independent network executing a consensus protocol security (*e.g.*, BC with its own set of validators or miners)
L2	Layer 2	An overlay network that relies on other BC for the security (*e.g.*, immutability)
DeFi	Decentralized Finance	Financial applications that rely on BC for their security and integrity (*cf.*, Tab. 3)
CeFi	Centralized Finance	Financial applications that are hosted outside of BC, but involve BC in their business model (*cf.*, Tab. 3)
TradFi	Traditional Finance	The traditional financial system, unrelated to BC or DeFi (*cf.*, Tab. 3)
CEX	Centralized Exchange	Exchange that allows swapping fiat currency to cryptocurrencies and tokens
DEX	Decentralize Exchange	Exchange that operates entirely on the BC and allows to swap cryptocurrencies and tokens
AMM	Automated Market Model	Mathematical formula used to determine the exchange rate between tokens

2.1 Blockchain

A blockchain (BC) is a computer network that maintains a distributed database. This network only allows appending new transactions and forbids deleting or updating any transactions. Transactions are grouped into blocks and block order in the database is maintained, and verifiable, using cryptographic hash functions and digital signatures. BCs can be classified as permissioned and permissionless. Permissionless BCs are open (public) networks accessible by all. Permissioned BCs can be accessed only by external parties recognized by the system administrator [99]. For this paper, we define a BC as a public - trustless and permissionless - distributed ledger. Consequently, DeFi applications are also trustless - do not depend on any centralized entity - and permissionless - open to use by any BC wallet.

A network *consensus mechanism* defines the rules, for what constitutes a legitimate transaction and block. The dominant consensus mechanisms are proof-of-work (PoW) and proof-of-stake (PoS). The consensus mechanism economically incentives network participants - miners in PoW and validators in PoS blockchains - to promote network security. In PoW BCs, miners, to append a new block to the ledger, are required to solve the cryptographic puzzle in a process that requires electricity[114]. In PoS BCs, validators, to append a new block, must pledge ("stake") a certain amount of native tokens of the underlying BC. In exchange for appending a new block, validators receive a reward. In case of malicious behavior, validators are punished, in a process called *slashing* by a fee from the staked tokens. *Staking* is the process of providing BC native tokens to validators to participate in profits from the block rewards. By design, PoS blockchain requires validators and *stakers* to freeze their tokens for a period of time that may vary from days to years. *Gas fee* is paid to miners and validators for appending the transaction to the BC.

2.2 Layer-1 vs Layer-2

According to *blockchain trilemma*, the blockchain network can only prioritize two features between three: decentralization, security or scalability[75]. As security is an absolute requirement and decentralization is a promise of BC, scalability has remained a challenge, resulting in low transaction rates and high transaction processing latencies. *Layer-1* is the blockchain with its own, independent trust assumptions: the network of nodes, and consensus mechanism [84]. Layer-1 solutions target the improvements of the core elements of blockchain design, *e.g.*, block data, consensus mechanisms, or sharding the network. *Layer-2* solutions aim to scale the BC without modifying the underlying trust assumptions. Layer-2 protocols are built on top of layer-1 BCs. *Roll-ups* are the most common scaling solutions for DeFi protocols deployed to Ethereum[17]. They aim to reduce the load of the main chain by taking the transaction execution off the chain in batches and bundling them together for on-chain verification. Depending on the verification process, roll-ups can be divided into two groups: optimistic roll-ups, *e.g.*, Arbitrum[6], Optimism[39], and zero-knowledge (zk) roll-ups, *e.g.*, zkSync[25], starknet[22]. Table 2 summarizes critical definitions and abbreviations used in the paper.

2.3 DeFi vs CeFi

Decentralized Fiance (DeFi) is a general term that refers to all financial applications that rely on blockchain for their security and integrity. *Centralized Finance* (CeFi) describes the financial applications that are hosted outside of BC, but involve digital assets in their business. An example of such CeFi application is *centralized exchange* CEX *e.g.*, Binance [7] or Kraken[16], which allows exchanging the fiat currency (USD, EUR, CHF, etc) into cryptocurrencies. Those cryptocurrencies purchased at CEX can be further transferred to the crypto wallet on BC or remain in the custody of CEX. *Traditional Finance* (TradFi) refers to the traditional finance system that does not involve digital asset-related activities or products.

Table 3. Comparison of DeFi, CeFi, and TradFi [119],
○= Yes, ●= No, ◐= depends on the specific protocol

	BC Related	BC Settlement	Non Custodial	DAO Governance
DeFi	●	●	●	◐
CeFi	●	◐	○	○
TradFi	○	○	○	○

Boundaries between CeFi and DeFi are not always clear and the work [118] provides a structured methodology for differentiation. If the user is not in control of the tokens (*e.g.*, does not retain custody, cannot transfer the assets without the financial intermediary), the financial application classifies as CeFi. CeFi applications may operate with BC settlement or not. In November 2022, FTX, the second largest CEX in the world collapsed, leaving its customers without access to their cryptocurrencies[14]. If the user is in control of the digital assets, but the financial intermediary can censor the transaction, such application classifies as CeFi with the BC settlement. Examples include USDT and USDC - stablecoins that are backed by fiat money. The issuer of USDT destroyed 44M of blacklisted USDT[118]. DeFi protocols can be further classified by their governance model: centrally governed and DAO. Centrally governed DeFi protocols are governed from outside of the BC. In the DAO model, decisions are made by holders of governance tokens via voting.

Table 4. DeFi stakeholders, based on [58, 95]

Stakeholder	Role	Incentive
Service Customers	Interacts with the DeFi protocol	Profit, credit, liquidity mining
Liquidity Providers	Provides capital to DeFi protocols to ensure sufficient liquidity for financial services	Protocol fee profit participation, liquidity mining
Arbitrageurs	Eliminate the market inefficiencies between different DeFi protocols	Arbitrage Profits
Governance Users	Design, develop and maintain the DeFi protocol	Governance token appreciation

2.4 DeFi Stakeholders

Blockchain is a decentralized network, in which its stakeholders (nodes, validators) operate in *peer-to-peer model*, without any system administrators or centralized entity. DeFi protocols are smart contracts deployed to BC and DeFi users interact with those smart contracts. *Liquidity pool* is a smart contract that holds tokens deposited in a DeFi protocol by *liquidity providers* (LPs) [93] in order to facilitate financial services (*e.g.*, borrowing, token swapping). LPs thus fulfill a similar role to that of market makers in TradFi. DeFi stakeholders - service customers of DeFi protocols, LPs, arbitrageurs, and governance users (*e.g.*, founders, developers, investors) interact with the liquidity pools, operating in *peer-to-pool* model[58, 95]. DeFi stakeholders have different incentives and roles in DeFi, and consequently are exposed to various risks. The table 4 summarizes the DeFi stakeholders and risk analysis follows in the further sections.

Liquidity mining is a process of interacting with a DeFi protocol to collect its governance tokens or other rewards. The process is not related to mining in PoW BCs[93]. The prime objective of liquidity mining is to attract new users and LPs to DeFi protocols.

2.5 DeFi Value Proposition

As summarized in figure 1, DeFi protocols provide various value propositions to users - trading, lending, asset management and BC interoperability - and extend the initial objective of BCs: a decentralized payment system and store of value[114].

2.5.1 Trading. *Decentralized Exchanges* (DEXs), *e.g.*, UniSwap[52], allow DeFi users to swap between two tokens with the settlement occurring directly on the blockchain. Compared to Centralized Exchanges (CEXs), DEXs offer higher diversity of tradeable tokens and higher security of transactions. *DEX Aggregators*, *e.g.*, 0x[131], are DeFi protocols that collect exchange rates from various DEXs and the best execution options. The major risks of DEXs for DeFi users include slippage risk and MEV attacks. Slippage cost refers to the difference between the price quoted by the DEX and the executed price. The MEV (maximum extractable values) attacks are executed by miners or validators that re-order transactions for their profits, leaving the DeFi user with higher slippage costs.

2.5.2 Lending. Lending Protocols allow DeFi users to borrow tokens against collateral. *Interest rate protocols*, such as Compound[121], and Aave[2], create pools of tokens that can be lent and borrowed, thus they are also referred to as protocols for loanable funds. Depending on token supply and demand, the protocols automatically adjust interest rates. Another mechanism for on-chain lending is provided by *crypto-backed stablecoin* protocols, such as MakerDAO[108], and Liquity[35]. Stablecoins are tokens that hold their value pegged to the fiat currency, typically US dollar. In order to hold the peg to the reference value, crypto-backed stablecoin holds collateral of other crypto tokens to ensure that the circulating token has a redemption value. Any DeFi user can borrow stablecoins directly from the stablecoin protocols by proving crypto collateral. In such a case, a collateralized debt position (CDP) is created. The major risk of DeFi lending is the liquidation risk of the underlying collateral. The liquidation occurs when the collateral value descends behind the minimum over-collateralization threshold. The dep-peg risk refers to the circumstances when stablecoin fails to hold the reference value.

2.5.3 Asset Management. The DeFi Protocols in asset management allow DeFi users to generate additional yield on their tokens. *Yield farming* is a process of providing liquidity capital to DeFi protocols in exchange for fee participation rewards[133][134], similarly to market making in traditional finance. *Impermanent Loss* is a major economic risk associated with yield farming, caused by the high volatility of the crypto-tokens. *Liquid staking* refers to DeFi protocols that accumulate rewards from staking without locking the tokens. The staking process at PoS BCs supports the security of the underlying BC, but requires locking staked tokens for a defined period to collect the staking reward. *De-Peg risks* is the major risk related to liquid staking.

2.5.4 Blockchain Interoperability. Blockchains, by design, are siloed computer networks that do not communicate with each other. Bridges allow DeFi users to *(i)* transfer tokens between blockchains without employing any CEX, and *(ii)* and to hold exposure to the tokens from other blockchains, *e.g.*, exposure to Bitcoin on the Ethereum chain. To achieve this objective, bridge protocols create *wrapped tokens* - tokens with the value pegged to the tokens at other blockchains, *e.g.*, wrapped Bitcoin[50] on the Ethereum chain. Wrapped tokens are exposed to the de-peg risk.

3 DEFI PROTOCOLS

This section presents the mechanics behind the major DeFi categories: decentralized exchanges (DEXs) and their aggregators, lending protocols (also referred to as interest-rate protocols or protocols for loanable funds), crypto-backed stablecoins (also referred to as collateralized debt positions) with other classes of stablecoins, bridges (wrapped tokens) and liquid staking protocol (liquid staking tokens/derivatives).

3.1 Decentralized Exchanges

Decentralized Exchange (DEX) is a class of DeFi Protocols that facilitates the non-custodial swap of digital assets with on-chain settlement [132] [104] and is the largest in terms of TVL class of DeFi protocols[12]. Depending on the price discovery mechanism, DEXs can be further classified into one of two models: *central limit order book* (CLOB)-based DEX and liquidity pool based DEX, [95]. While most DEXs support swap of digital assets within one BC, the new class of DEXs - *multi-chain DEX* - support swap of digital asset between BCs [132].

The first implementations of DEXs were based on CLOB, similar to the exchange mechanism in *centralized exchanges* (CEX) and traditional finance. CLOB matches buyers with sellers - *peer-to-peer model* - and the swap price is the last matched price. This design however proved to be infeasible and expensive at scale on BC. Storing even a small amount of data on BC is expensive, which, combined with complex order-matching algorithms, leads to excessive fees [95]. The current CLOB DEX implementations solve this issue by implementing its own Layer1 or Layer2 BC, *e.g.*, dYdX[96], or moving the storage of CLOB outside of BC (off-chain order book), *e.g.*, 0x Order Book[131]. Off-chain order-book changes may be classified as belonging to CeFi with on-chain settlement, rather than to DeFi [118].

In contrast to the CLOB-based DEX, liquidity pool-based DEX does not store bids and ask orders at all. Instead, it employs reserves of two or more tokens, called *liquidity pools*. Liquidity pools provide counterparty tokens to swap against - *peer-to-pool/peer-to-(smart) contract* model - and the swap price is determined algorithmically by the pricing rules called *automated market maker* (AMM). Liquidity pool-based DEX, also referred as *AMM-based DEX*, became dominant in DeFi for several of reasons: 1) liquidity pools can be created easily for any pairs of digital asset, including new and minor tokens 2) anyone can provide capital to liquidity pools 3) liquidity pools and AMMs are more efficient compared to CLOB [132].

Depositors that seed liquidity pools with capital in return for swap fees, are referred to as *liquidity providers* (LPs). The trading fees are distributed to LPs proportionally to the capital provided to the liquidity pool. The process of providing liquidity to the DEX, or other class of DeFi protocols, in return for profit, is called *yield farming* and is similar to market making in traditional finance. Market makers in traditional finance provide both bid and ask orders from their own book and make a profit from the bid-ask spread.

Arbitrageurs buy and sell the same asset at various exchanges to profit from the price differences. Even though small liquidity pools may lead to significant fluctuations in the swap prices, arbitrageurs ensure that the swap price at the DEX is at parity with the market. The next section presents the most used AMM.

3.1.1 Automated Market Maker (AMM).

: An AMM is a mathematical function that algorithmically determines the swap price between tokens in the liquidity pool via a scoring rule. The most popular scoring rule is the logarithmic market scoring rule from Robin Hanson [87]. In recent years new AMMs emerged in the field of decentralized finance, the *constant function market makers* (CFMM). A CFMM is a trading function $\varphi : \mathbb{R}^n_+ \rightarrow \mathbb{R}$, often called conservation function or reserve curve, that maps the token reserves R_i to an constant k, often referred to as invariant. Therefore, the conservation function is a relational function between an invariant k and token reserves R_i. In addition, the conservation function is always concave, nonnegative, and nondecreasing[54] and the spot price is given by the slope of the curve at the current reserves level $p_{R_i,R_j} = \frac{\partial\varphi}{\partial R_i} / \frac{\partial\varphi}{\partial R_j}$. If the liquidity pool contains only two tokens, the two reserves are typically referred to as x and y. Various AMM formulas are currently in use by DEXs. The most popular AMMs fall into the category of *constant function market making* (CFMM), and include the following *reserve curves* [115, 134]:

3.1.2 Constant Sum Market Maker.

$$x + y = k$$

The constant Sum Market Maker (CSMM) is the simplest form of an AMM. It sums over all reserves and the sum is defined as the invariant k, resulting in a swap exchange rate of 1. This means that a small deviation from the price will lead to arbitrage and one token reserve will be completely drained. Therefore, this AMM is unsuitable as a standalone solution for DEXs.

3.1.3 Constant Product Market Maker.

$$x \cdot y = k$$

The Constant Product Market Maker (CPMM) is the most known DeFi AMM and became popular through its use in Uniswap v2. The function defines the interaction of two token reserves and the invariant by a simple product of the two reserves. In a swap operation, this always remains the constant k, so Δx of x can be swapped for Δy of y, resulting in $(x + \Delta_x)(y - \Delta_y) = k$ and a spot exchange rate of $p_{x,y} = \frac{x}{y}$.

3.1.4 Constant Mean Market Maker.

$$\prod_{k=1}^{n} R_i^{w_i}$$

The Constant Mean Market Maker (CMMM) is the extension of the CPMM. This AMM includes more than 2 token reserves R_i and weights them with w_k and was first introduced by Balancer in 2019[109]. It results in an spot exchange rate of $p_{i,j} = \frac{R_i \cdot w_j}{R_j \cdot w_i}$.

3.1.5 Stableswap Invariant.

$$A \cdot n^n \cdot \sum_{i=1}^{n} R_i + k = A \cdot k \cdot n^n + \frac{k^{n+1}}{n^n \prod_{i=1}^{n} R_i}$$

The Stableswap Invariant[80], often also referred to as Curve AMM, mixes the behavior of the CPMM and CSMM and is mostly meant for Stablecoins. The conservation function includes a leverage/amplification coefficient $A \in \mathbb{R}_+$ which describes how strongly the function follows a CPMM or CSMM. For $A \longrightarrow 0$ the function behaves like the CPMM, for $A \longrightarrow \infty$ like the CSMM. The spot exchange price is given by

$$p_{i,j} = \frac{R_i \cdot (A \cdot R_j \cdot \prod_{l=1}^{n} R_l + \frac{k^{n+1}}{n^n})}{R_j \cdot (A \cdot R_i \cdot \prod_{l=1}^{n} R_l + \frac{k^{n+1}}{n^n})}$$

Figure 2 gives a graphical representation for each of these conservation functions.

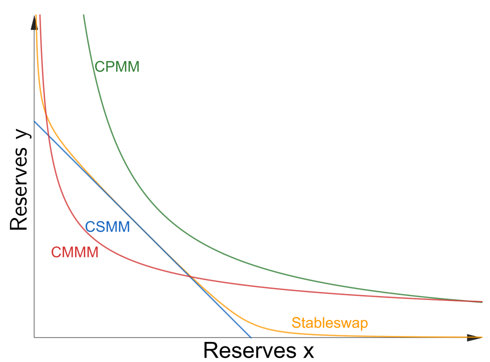

Fig. 2. Graphical representation of the AMM conservation functions for two tokens with CMMM weights $w_x = 1$, $w_y = 2$ and a stableswap leverage coefficient of $A = 10$

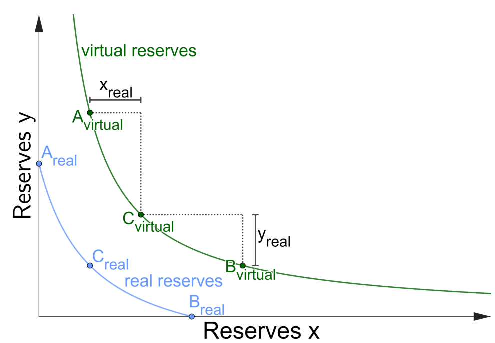

Fig. 3. Concentrated liquidity and virtual reserves, based on [52]

3.1.6 Concentrated Liquidity. Concentrated Liquidity AMM (CLAMM) is an enhancement to AMM that increases capital efficiency in liquidity pools[4]. It was first proposed by UniSwap v3[52] and allows LPs to define the price range, for which liquidity is provided. LPs profit from swap fees when the price remains in the defined interval[4]. Consequently, there is a trade-off between larger (less risky, less efficient) intervals and (more risky, efficient) smaller intervals. Consider a liquidity range interval $[p_a, p_b]$ around the current spot price p_c, then x_{real} and y_{real} denotes the position real reserves. As soon as the price moves outside the range, a reserve of the liquidity provider is completely used up. Consequently, CLAMM requires active management and accepting higher financial risk, compared to AMM [90]. The liquidity provided is measured by the invariant L, which equals $\sqrt{k}$ in the CPMM setting. The real reserves of a position are described by a translation of the standard CPMM formula:

$$(x + \frac{L}{\sqrt{p_b}})(y + L \cdot \sqrt{p_a}) = L^2$$

Therefore, the spot exchange rate is identical to the CPMM case ($p_{x,y} = \frac{x}{y}$). Figure 3 shows the real and virtual reserves and the shift in the curve.

3.1.7 Risks. The major risks for the participants of AMM-based DEX include 1) slippage cost for traders, 2) impermanent loss for LPs 2) and transaction-ordering attacks. Those risks come additionally to the explicit costs of DEX - swap fees and gas fees. Transaction reordering, or MEV-attack, is further discussed in this work, as it poses a risk for many classes of DeFi protocol.

Slippage is the difference between the spot price, presented by the DEX, and the actually realized swap price. The difference is caused by AMM and the reserve curves. The realized price diverges from the spot price for larger swapped amounts and the divergence is further amplified by smaller liquidity pools. Swap transactions at AMM DEX are set with the *slippage tolerance*. Slippage tolerance restricts losses from the MEV attacks.

Impermanent loss is the difference between holding tokens in the wallet and allocating them to the liquidity pool. Every swap transaction alters the liquidity pool's composition, increasing the value-losing token's reserve. Impermanent loss can be eliminated by providing liquidity for stablecoin pairs [106] or mitigated by including stablecoin in the token pair [91]. The impact of impermanent loss has been widely studied across various AMMs[98, 128].

3.2 DEX Aggregators

The rise of DEXs on the same blockchain provides more competitive options for DeFi users, but also leads to fragmented liquidity among various AMM pools and requires arbitrageurs for price synchronization. DEX arbitrage and trade routing can be done efficiently on-chain within smart contracts[138]. *DEX aggregators* provide the optimal routing of trade orders across multiple DEXs [55]. Around 21.5% of DEX volume is initiated by aggregators[30].

3.3 Interest-rate protocols

Interest rate protocols, also referred to as lending protocols, facilitate borrowing and lending of digital assets [74, 102, 122]. Lenders - liquidity providers (LPs), provide capital to the protocol's pool - *lending pool*, from which it is distributed to borrowers. The interest rate mechanism seeks for equilibrium between the demand and supply of the capital [86]. DeFi lending protocols are thus not decentralized peer-to-peer lending strategies with direct relations between borrowers and lenders. Instead, they operates in a peer-to-pool model that avoids individual matching. Thus, the LP has a diversified risk across all borrowers of this pool and a utilization rate U_t can be calculated at time t. The utilization rate is the ratio of the toal amount borrowed at time t to the liquidity at time t, so $U_t := \frac{B_t}{L_t}$[2]. Liquidity is provided to the lending pool managed by the smart contract of the protocol. Thus, the DeFi lending protocols are also referred to as *protocols for loanable funds* (PFLs)[86]. Unlike in traditional banking, DeFi participants are anonymous, preventing assessing the risk of borrowers [57]. To mitigate the default risk, DeFi interest-rate protocols are over-collateralized: a borrower is required to post collateral covering the value of the debt and the value of the posted collateral must exceed the value of the debt. If the value of the locked collateral falls below the *liquidation threshold*, the collateral can be purchased for a discounted value and the debt position is closed [117]. DeFi loans typically have unlimited maturities and the interest rate is variable, so it fluctuate from the opening of the loan position and is recalculated constantly. Interest is accrued per second and paid per block[86]. Based on the interest rate model, the lending protocol can be classified into three major classes: linear, non-linear and kinked rates.

3.3.1 *Linear rates.*

In a linear model, interest is a linear function of the utilization rate. For the borrower, the variable interest rate $i_{b,t}$ results from:

$$i_{b,t} = \alpha + \beta \cdot U_t,$$

where α is a constant and β describes how the interest rate behaves for changes in the utility rate. The lender receives a lower saving interest rate $i_{s,t} = i_{b,t} \cdot U_t$ which also depends on U_t [86] For example, the first version of Compound used this model, but it has since been replaced by a kinked rate model [121].

3.3.2 *Non-linear rates.*

In a non-linear model interest rates increase with an increasing rate as the utilization grows. The interest rate structure was used in the old dYdX layer 1 and DDEX[?], where the borrowing interest rate is given by

$$i_{b,t} = (\alpha \cdot U_t) + (\beta \cdot U_t^n) + (\gamma \cdot U_t^m),$$

with $n, m \in \mathbb{R}$ The saving interest rate $i_{s,t} = (1 - \lambda) \cdot i_{b,t} \cdot U_t$ includes a reserve factor λ, which puts aside a fraction of the earned interest rate for more illiquid periods [86].

3.3.3 *Kinked rates.*

In the kinked model, interest rates increase more sharply above a certain (optimal) utilization rate U_{opt}, i.e. there is a kink in the interest rate slope. This is used in the current version of AAVE [2] and Compound [121]. The borrower rate can be characterized by

$$i_{b,t} = \begin{cases} \alpha + \beta U_t & \text{, if } U \leq U_{opt} \\ \alpha + \beta U_{opt} + \gamma (U - U_{opt}) & \text{, if } U > U_{opt} \end{cases},$$

where β describes the interest rate slope up to the optimal utilization rate and $\gamma >> \beta$ the slope for a higher utilization.

3.3.4 *Total value locked.*

(TVL) for the interest-rate protocols represents the value of tokens hold by the protocols, which is the difference between the supplied and borrowed funds, and the value of the collateral. The major interest-rate protocols, Aave[2] and Compound[121] with USD 3.37bn and USD 1.64b TVL respectively[11], are examples of DeFi interest-rate protocol with kinked rates. The exchange dYdX[96] implemented non-linear rates to offer loans to its users.

3.3.5 *Liquidation Risk.*

In case the value of the collateral drops below the liquidation threshold, the borrower automatically defaults on its debts, and the collateral is sold off with the discount, in the process referred to as *liquidation* [117]. *Zero liquidation loans* (ZLLs) are the novel approach to mitigate the DeFi liquidation risk by defining the loan tenor (duration) [124]. The ZLLs protocols such as [49, 116, 123] offer the borrower an option (but not obligation) to reclaim the collateral upon the repayment of the debt and the fees. If a borrower does not repay the debt before the ZLL expires, LPs gain the right to liquidate the collateral.

3.3.6 *Flash loans.*

Flash loans are a special class of DeFi loans that allow to borrow without a collateral during a single transaction. The debt is protected by the atomicitiy enforced by smart contracts and blockchain: if the loan is not repaid with the interest withing the block, the whole transaction with flash loan is reversed[2]. Flash loans increase the efficiency of DeFi lending by allowing to purchase any digital assets that can be used as a collateral, including NTF [60], as presented in the algorithm 1. The platforms [28, 38] accept certain NFTs as a collateral.

Algorithm 1 Mortgaging NFT[60]

1: **Flash loan** x_i of token i to an address holding y_i of token i
2: **Purchase** the NFT for $x_i + y_i$ of token i
3: **Deposit** the NFT as collateral, with borrowing power of $z_i > x_i$
4: **Borrow** x_i of token i to repay the flash loan

3.4 Yield Farming

Yield farming is the process of earning income by providing tokens to DeFi protocols for market-making purposes[77] while token providers are referred to as "yield farmers". The yield farming protocols automate the yield farming process and can be perceived as algorithmic asset managers that execute the pre-programmed strategy [135]. Technically, yield farming protocols are aggregators that automatically deploy tokens to other DeFi protocols [133] and are consequently also referred to as "yield aggregators". The income from yield farming activities might be amplified by the participation rewards from DeFi protocols, to which tokens are provided. Such participation rewards are typically offered as governance tokens. Yield farming protocols may charge management and performance fees.

3.4.1 Simple lending. In this strategy, the tokens are provided to the interest rate protocols (*e.g.*, Compound, Aave), which lend them to the borrower. The token provider (supplier) accrues the supply interest. Given the over-collateralization required by the interest rate protocols from the borrowers, the strategy is low-risk and losses occur only under extreme market conditions. Based on the source of yield, the protocol strategies can be divided into the three groups[77]: *(i)* simple lending, *(ii)* leverage borrow, *(iii)* liquidity provision.

Algorithm 2 Simple Lending Yield Farming Strategy[77]

1: **Deposit** token i in a lending protocol
2: **Accrue** interest of token i and collect potential other rewards, e.g. native tokens z over time
3: **Withdraw** deposits with accrued supply interest and other rewards

3.4.2 Leverage borrow. This strategy aims to maximize the yield from lending by leveraging spirals: tokens deposited in the interest rates protocols are used as collateral to borrow new tokens. The new tokens are deposited in the interest rate protocol and used again as collateral to borrow. The process is repeated multiple times. A high degree of leverage can amplify both profits and losses.

Algorithm 3 Leverage Borrow Yield Farming Strategy[77]

1: **Deposit** token i in a lending protocol
2: **Borrow** token j with the deposits as collateral
3: **Deposit** borrowed token j as collateral
4: **Repeat** steps 2–3 multiple times
5: **Accrue** interest and collect potential other rewards, e.g. native tokens z over time.
6: **Swap** the native tokens into the assets borrowed
7: **Repay** loans with accrued borrow interest
8: **Withdraw** deposits with accrued supply interest and other rewards

3.4.3 Liquidity provision. With this strategy, tokens are provided to liquidity pools (LPs) of AMM DEXs to earn income in terms of transaction fees charged by DEXs.

Algorithm 4 Liquidity Provisions Yield Farming Strategy[77]

1: **Provide** token i, j as liquidity in an AMM pool
2: **Collect** transaction fees in token i, j and potential other rewards, e.g. native tokens over time
3: **Withdraw** liquidity, accrued fee participation and other rewards

3.5 Stablecoins

Stablecoins are tokens, which value is pegged to a fiat currency, typically 1 USD. They provide a way for DeFi users to mitigate the volatility of other digital assets, without the need to transfer liquidity outside the blockchain[113]. Holders of stablecoins are exposed to the *de-peg risk* - the risk that the stablecoin will not hold the peg to the target value. Stablecoin protocols employ various mechanisms to hold the peg and depending on the collateral type, they can be classified into three categories: *(i)* fiat-backed, *(ii)* crypto-backed, and *(iii)* algorithmic stablecoins[64]. The collateral ensures that the circulating token has redemption value.

3.5.1 Fiat-backed stablecoins. Fiat-backed stablecoins utilize collateral of a fiat currency or commodity outside the blockchain, thus, they are also referred to as off-chain collateralized stablecoins. Given the collateral held outside of the blockchain and regulatory restrictions, fiat-backed stablecoins are not considered to be part of DeFi, but CeFi[118].

3.5.2 Crypto-backed stablecoins. Crypto-backed stablecoins maintain the collateral of other crypto tokens on the blockchain. They are also referred to as on-chain collateralized or over-collateralized stablecoins. Over-collateralization implies that the total value of the collateral is higher than the total value of the circulating stablecoin supply and is applied to protect the peg against the volatility of crypto tokens and related sudden price changes of the collateral. Any DeFi user can mint the new stablecoins, by providing sufficiently over-collateralized collateral. The process can also be seen as borrowing stablecoins with crypto tokens locked as collateral.

The borrowing process from the stablecoin protocol is initiated by depositing crypto tokens as collateral. The collateralized debt position (CDP) is created. In exchange for the deposit, the depositor receives stablecoins, which are minted by the stablecoin protocol. Once the borrower returns the stablecoins, the CDP is closed: the tokens used as collateral are returned to the borrower and the stablecoin protocol burns returned stablecoins.

3.5.3 Algorithmic stablecoins. Algorithmic stablecoins maintain their price peg by managing (shrinking or expanding) the supply of the token. As they do not employ any collateral, algorithmic stablecoins are also referred to as non-collateralized stablecoins. There are two main types of algorithmic stablecoins: seigniorage and rebasing [112]. The rebasing coins, such as Ampleforth[27], adjust the balance of coins in the users account, allowing the coin place to fluctuate. Consequently, some literature does not consider the rebasing coins to be stablecoins. Seigniorage stablecoin protocols, such as Terra[46], make arbitrary decisions on the minting and burning of the coins in the supply in order to maintain the peg.

3.5.4 Risks. De-peg risk - the major risk related to the stablecoin - references to the circumstances, in which the stablecoin loses the peg to the target value. The *stablecoin trilemma* indicates that any stablecoin design poses limitations related to one of the following risks: *(i)* hazards of the operating entity, *(ii)* exposure to the external market risk, or *(iii)* limited coin supply [97]. The major risk of DeFi lending, including borrowing from the stablecoin protocol, is the *liquidation risk* - the risk of liquidating the CDP with the locked collateral. The liquidation occurs when the collateral value descends below the minimum over-collateralization threshold. Borrowers are required to monitor the over-collateralization level of their CDP to avoid liquidations. The automatic detection of the CDP that needs to be liquidated and initiating the liquidations are not trivial on the blockchain. Smart contacts, by their design, need to be triggered externally and do not support automation.

3.5.5 CBDC. Stablecoins are typically discussed in relation to the topic of *Central Bank Digital Currencies* (CBDC), as a potential implementation of CBDCs. CBDC is a form of digital cash issued by the central bank. CBDC may be but does not have to be related to the BC or DLT, as it is the issuer - the central bank - of digital cash that is a defining factor of CBDC. CBDCs are further classified as wholesales or retail[70]. The potential CBDC implementation may be decentralized or centralized, and may not relay on DLT[70].

3.6 Liquid Staking

Synthetic assets in DeFi are analogous to derivatives in traditional finance (TradFi) [120]. In TradFi, derivatives allow traders to profit from price movements of stocks and bonds they do not own[82]. Synthetic assets are cryptocurrency tokens that track the price fluctuations of the target asset[59, 120]. *Liquid staking* is a synthetic asset that tracks the value of the tokens staked in the proof-of-stake blockchains [126]. Liquid staking token is also referred to as *liquid staking derivative* (LSD) and the protocol minting it - *liquid staking protocol* (LSP). Staking is the process of locking tokens in validators for future profit [126]. Validators, network participants in PoS blockchain, put those tokens at stake when appending new transactions to the blockchain. For each correctly appended transaction, the validator receives a reward, which is redistributed between token providers (*stakers*). The dishonest behavior is punished by a reduction in the staked tokens (*slashing*). By design, the staked tokens are locked for a specific period, making it impossible to use them in other applications, reducing the token liquidity. By tracking the value of the staked assets, liquid staking tokens allow participating in the staking benefits. Unlike the staked tokens, liquid staking tokens are not locked but are freely traded and used in other DeFi applications. There are three standard LSD models [26, 126]: *(i)* rebasing tokens, *(ii)* reward-bearing tokens, and *(iii)* dual-token model.

3.6.1 Rebase Tokens. In this model, the LSP tokens keep the 1:1 peg to the staked tokens and inflate the supply to reflect the yield earned from staking. The most popular rebase LSD is stETH, minted by Lido Protocol[3]. Lido every day increases the balance of all stETH holders to reflect the additional ETH earned as staking rewards. The advantage of rebasing LSP is its simple functionality. The disadvantage is lack of the compatibility with other DeFi protocols, *e.g.*, DEXs or interest rate protocols.

3.6.2 Reward-bearing Token. In this model, the token does not maintain 1:1 peg to the staked native token, but increases its value to reflect the staking yields. Consequently, the reward-bearing token grows in value, related to the underlying native token. Reward-bearing tokens provide better compatibility with DeFi protocols. Most of LSPs follow this design: RocketPool[43], Marinade [36], CoinBase Wrapped Staking [9].

3.6.3 Dual Token. In the dual token model the asset token is split from the staking revenue it yields. For example, StakeWise [45] operates in the two token model: sETH2 - with 1:1 peg to ETH - and rETH2 - that reflects the earned staking rewards. The disadvantage of this model is the fragmented liquidity between two tokens.

3.6.4 Other Classifications. LSP also varies in the validator selection process: *(i)* whitelisting reputable validators, *(ii)* credentials-based, *(iii)* collateral. In the whitelisting model, LSP adds only trusted validators to the pool of validators, *e.g.*, Lido[3]. In the credential based-model, the pool of validators is permissionless upon fulling the minimum criteria.

Table 5. Overview of selected liquid staking tokens

Tokens	Protocol	Blockchain	Model
stETH	Lido	Ethereum	rebasing
wstETH	Lido	Ethereum	reward-bearing
mSol	Marinade	Solana	reward-bearing
rETH	Rocket Pool	Ethereum	reward-bearing
sETH2, rETH2	StakeWise	Ethereum	2 tokens

In the last model, LSP requires validators to post the collateral to guarantee its performance, *e.g.*, Rocket Pool[43]. The collateral model hampers the growth of the validator pools by increasing the minimum threshold for staking.

3.7 Bridges

DeFi ecosystem has evolved into a multi-chain ecosystem, but each BC is an independent and closed network by design. Miners and validators write transactions to the BC according to its consensus mechanism. Fragmentation of DeFi protocol and liquidity across various chains hinders DeFi growth and locks DeFi users to a single BC. Blockchain interoperability includes the transfer of digital assets, data, and smart contract instructions between BCs, which would allow DeFi users to utilize digital assets stored in one BC in DeFi protocols on another BC (*cross-chain collateral*) and execute low-cost and fast transaction of digital assets stored on less scalable BC (*efficiency and scalability*). For the DeFi protocols, blockchain interoperability would allow for more efficient utilization of liquidity, locked across various BCs [5]. While Polkadot[19] and Cosmos[10] provide frameworks to develop interoperable BCs, *e.g.*, by sharding, they do not facilitate cross-chain communication with networks outside of their BC-framework[136]. Centralized exchanges (CEX) can be used to transfer digital assets across blockchains. However, given their centralized nature, they do not deliver on BC and DeFi promises of decentralization, permissionless and trustless.

Bridges enable the transfer of tokens between BCs [67, 88, 89, 100, 101, 111, 127, 130, 137], *e.g.*, Bitcoin and Ethereum, or between layers of the same BC, *e.g.*, Ethereum and its roll-ups. Based on the underlying architecture, bridges can be classified as *(i)* wrapped tokens, *(ii)* liquidity networks and *(iii)* hybrid solutions[136].

3.7.1 Wrapped Tokens. Wrapped tokens can be further classified as *(i)* External Validators and Federations *(ii)* Light Clients and Relayers [61].

3.7.2 Liquidity Networks. Most DEX supports the exchange of digital assets only within the same BC. What is more Atomic cross-chain swaps (ACCS) proved to be slow, inefficient, and costly[136].

3.7.3 Other Classifications. The full overview and classification of existing solutions present [18]. Another classification of bridges: *(i)* trusted *(ii)* Binded *(iii)* Insured *(iv)* Trust-less [61].

3.8 Perpetuals

Perpetual contracts or swaps in DeFi are derivatives that allows to speculate on the price of an underlying asset, such as cryptocurrencies, without actually owning the asset. Perpetual contracts resemble traditional futures or options, however, they do not have a fixed expiration date. This allows to hold their positions indefinitely, as long as the required collateral limits can be maintained. They were first introduced by the centralized exchange BitMEX in 2016 [53], requiring counter-parties and thus a matching mechanism (similar to CLOB).

While automatic liquidation of collateral is an essential mechanism for any perpetual, another fundamental DeFi mechanism is necessary to remove the need for counter-parties: virtual AMMs (vAMM). A vAMM can use the same mathematical principles as an AMM for price determination, which is required for automatic liquidation of collateral once the price hits a certain limit, *e.g.*, if the price shifts to your disadvantage, you may require to provide more collateral. In case your collateral limit is reached (also depends on your leverage factor), your collateral will be automatically liquidated.

4 DEFI PROTOCOL TAXONOMY

DeFi protocols provide various financial services and can be classified based on the value proposition to users: trading, lending, asset management, and BC interoperability, as depicted in table 1. This section proposes novel classification frameworks. First, we separate the token classification from the taxonomy of DeFi protocols. Next, we classify DeFi protocols in two dimensions - algorithm and network (blockchain) architecture. *Taxonomy of DeFi algorithms* is based on the technical and economical design of DeFi protocols. The DeFi algorithm determines how financial services are performed on the BC and to which risks DeFi agents are exposed. *Taxonomy of DeFi network architecture*, refers to the number of BCs, on which the protocol operates and the coordination of its work across those BCs. The network architecture implies the infrastructure risks. The classifications within those two domains, as well as token taxonomy, are straightforward, as the values assigned to each dimension can be traced from the DeFi protocols' implementations and whitepapers that are publicly available.

4.1 Token Taxonomy

There exist multiple extensive token classification frameworks *e.g.*, [56, 58, 83]. Based on the work of [95] and [120] we select the two token classification domains: "technology" and "underlying value". Token technology comprises two dimensions: BC native tokens (*e.g.*, Bitcoin, Ethereum, Solana) and smart-contract-enabled tokens. The "underlying value" dimension comprises three dimensions: asset-backed, network value, and share-like[83]. Asset-backed tokens comprise tokens linked to other assets (*e.g.*, tokens confirming participation in liquidity pools), derivative tokens with other tokens as underlying (*e.g.*, stablecoins, liquid staking tokens), and real-world assets - tokenized assets from the real world. The proposed taxonomy, which combines and simplifies, other classification frameworks [83], emphasizes the risk of the token holder. This token classification can help to ensure a consistent view of the different token types and the underlying counterparty risk.

- **Native tokens**: Layer 1 BC's native tokens, *e.g.*, Bitcoin, and Ethereum.
- **Smart contract enabled tokens**: Tokens that are created within BC by smart contracts can be further divided into more granular dimensions:
  - **Network Value**: the underlying value is represented by the level of trust, *e.g.*, DAO governance tokens or other utility tokens for fee payments within the smart contract applications. Network-driven demand for the token may lead to an increase in value in case of limited supply.
  - **Asset-backed**
    - **On-chain Asset**: tokens tied to other on-chain tokens, *e.g.*, tokens representing participation in the liquidity pool in yield farming at AMM DEX,
    - **Synthetic Assets**: the underlying assets comprise another on-chain asset, *e.g.*, crypto-backed stablecoin, liquid staking token, or wrapped Bitcoin (for bridged tokens),

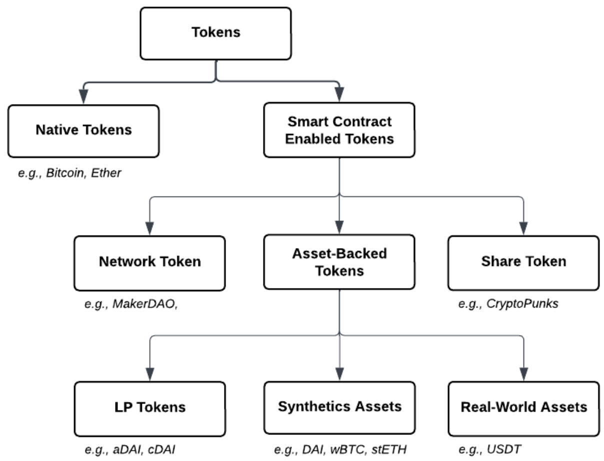

Fig. 4. Taxonomy of Tokens

  * **Real-World Assets**: the underlying value is tied to the tokenized real-world assets, *e.g.*, fiat money for fiat-backed stablecoins.
- **Share-like Tokens**: the underlying value is generated by the work of the token holder or 3rd party issuers *e.g.*, NFT representing art ownership.

4.2 Taxonomy of DeFi Algorithms

In the design domain, the DeFi protocol can be classified into three dimensions: *(i)* liquidity pools *(ii)* aggregators *(iii)* synthetic tokens, as presented in figure 6. The DeFi protocols are developed on the infrastructure layer of their underlying blockchains. The infrastructure layer includes native tokens or staking for proof-of-stake blockchains.

4.2.1 Liquidity Pools. Liquidity pool-based DeFi protocols match the demand and supply using token pools operated by smart contracts. The major DeFi liquidity pool-based protocols include decentralized exchanges (DEX) and interest-rate protocols. DEXs manage the supply and demand to swap tokens and, depending on the design, can be further classified as CLOB or AMM. There are various implementations of AMM, with concentrated liquidity being the latest improvement that increases the capital efficiency at AMM DEX. The interest-rate protocol manages the supply and demand for borrowing and lending and adjusts interest rates automatically. DeFi liquidity pools protocols and stablecoins are referred to as DeFi primitives [77].

4.2.2 Aggregators. The major aggregator protocols facilitate the operations of DEXs and include DEX aggregators and yield farming protocols. DEX aggregators allow DEX service customers - traders - to find the best execution rate, and sometimes offer protections again MEV attacks, by splitting the transaction into smaller ones. Yield farming protocols simplify and increase liquidity efficiency at AMM DEXs.

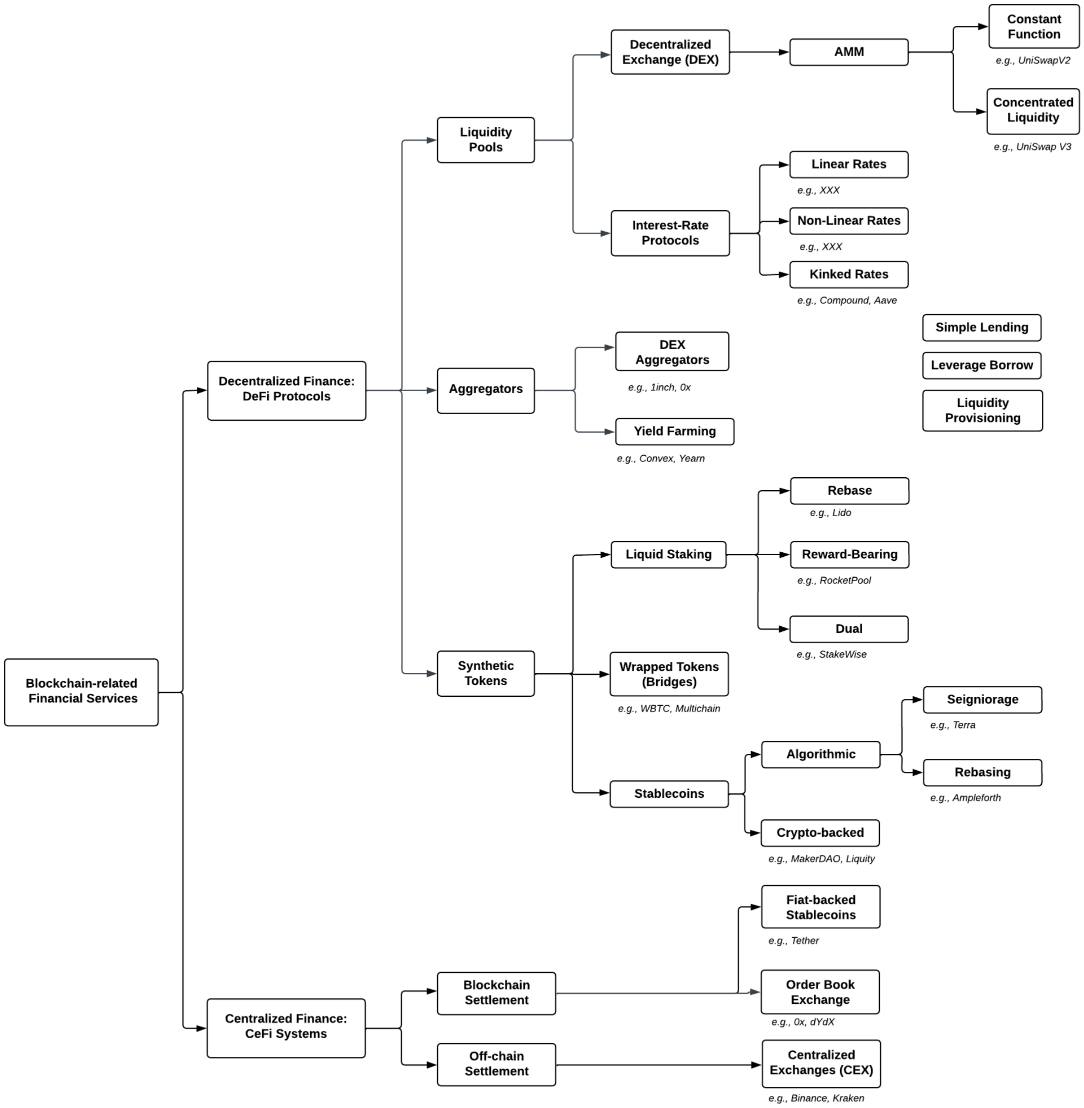

Fig. 5. Taxonomy of DeFi Algorithms

4.2.3 Synthetic Tokens. Synthetic tokens are similar to derivatives in traditional finance. They employ smart contracts to keep the peg to the target value, which might be fiat currency - stablecoins, tokens at other blockchains - wrapped tokens of bridges, and staked tokens - liquid staking. The inherent risk of all synthetic tokens is the re-peg risk.

4.3 Taxonomy of DeFi Network Architecture

This dimension indicated whether DeFi protocols operate on a single chain or across multiple chains. Protocols operating on multiple networks are in the early stage but they are expected to play a substantial role in DeFi in future[83].

4.3.1 Single Chain. Single-chain protocols are deployed and operate within one BC. Most of DeFi protocols operate on single chain[11].

4.3.2 Cross Chain. Cross-chain protocols that operate on multiple BCs and allow the value to cross between BCs. Those protocol utilize blockchain interoperability technologies so that the value stored on one BC is transformed into value at other BC[105]. The cross-chain protocols include bridges.

4.3.3 Multi Chain. Multi chain protocols are deployed to multiple blockchain, at which they operate independently. Contrary to cross-chain protocols, they do not utilize value cross BC, *e.g.*, cross-chain collateral. Example of DeFi protocols deployed to multiple chains include Aave[2], Uniswap[52].

4.3.4 App Chain. App-chain (or Solo-chain) is DeFi protocol that developed sovereign BC to optimise its services for the specific use cases. Depending on the security model of the sovereign BC, the app-chain can be classified at L1, L2 or L3. Cosmos[10] provides SDK for the development of L1 appchain. Polkadot[19] simplifies the development of L2 appchains, called parachains by Polkadot. Starknet[22] and zkSync[25] are zero-knowlege rollups (L2) on top of Ethereum and offer possibilities to built furhter L3.

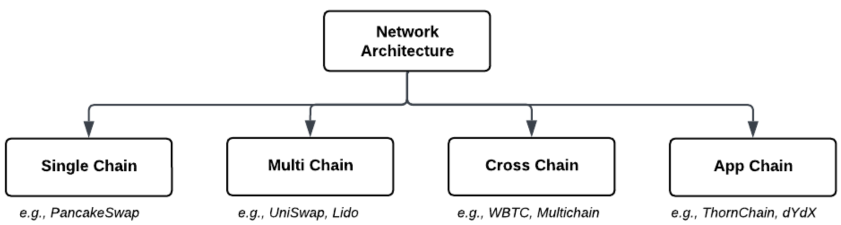

Fig. 6. Taxonomy of DeFi Network Architecture

5 DEFI RISK AND ATTACK CLASSIFICATION

With the rapid increase of TVL, attacks on DeFi protocols have gradually emerged [71, 103, 139]. In *DeFi exploits*, attackers gain profits from the vulnerability in the technical or economic security of the DeFi protocols [132]. Table 6 summarizes DeFi risks and their impact on DeFi stakeholders - service customers, liquidity providers (LPs), arbitrageurs, or governance users. This section discusses these risks and attacks in detail. As yield farming protocols are aggregators that provide liquidity to other DeFi protocols, this category possesses risks of LPs.

The blockchain trilemma states that every BC can prioritize only two from three factors: decentralization, security, and scalability[75]. The limited scalability of the BC results in network congestion and high transaction fees. A lack of decentralization allows for transaction censorship. The security attacks target vulnerabilities in the technical (*e.g.*, smart contract, transaction re-ordering) or economical design (*e.g.*, oracle price manipulation) of DeFi protocols and BCs.

Table 6. DeFi Security Risk Overview.
C: Service Customers, LP: Liquidity Provider, A: Arbitrageur, G: Governance User
○= Not Vulnerable, ●= Vulnerable, ◐= Depends

	CLOB DEX			AMM DEX			Interenst-rate Protocols			Stablecoins		Bridges		Liquid Staking		
	C	LP	A	C	LP YF	A	C	LP	A	C	A	C	A	C	A	G
DeFi Risks																
Rug Pull Attack	●	●	●	●	●	●	○	○	○	○	○	○	○	○	○	○
Slippage	○	○	○	●	○	●	○	○	○	○	○	○	○	○	○	○
Impermanent Loss	○	○	○	○	◐	○	○	○	○	○	○	○	○	○	○	○
Liquidation	○	○	○	○	○	○	●	○	●	●	○	○	○	○	○	○
De-peg	○	◐	○	○	◐	○	○	◐	○	●	○	●	○	●	○	●
Infrastructure Attack																
MEV Attack	●	○	●	●	○	●	○	○	○	○	○	○	○	○	○	○
Blockchain Attack	◐	●	◐	◐	●	◐	●	●	◐	●	◐	◐	◐	●	◐	●
Middle Layer Attack																
Smart Contract Attack	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
Application Attack																
Oracle-Attack	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●

- **Limited Scalability** Scalability is the central issue for BC networks[84]. The limited scalability leads to network congestion and spikes in transaction costs. During the network congestion, transactions of arbitrageurs, and other DeFi stakeholders, may not be executed, leading to market inefficiency and information delays[95]. Layer-2 solutions aim to scale the underlying main chain at the cost of various security assumptions[84].
- **Centralization Risk** Centralization might affect both the BCs and the DeFi protocols. DeFi Protocols increasingly rely on governance tokens to decentralize the decision-making process[94]. Nonetheless, the distribution of governance tokens may remain concentrated within the small groups of colluding stakeholders, as the distribution of voting powers is a gradual process[129].
- **Security Threats** DeFi Protocols operate on BC as smart contracts and are prone to security attacks on various layers: BC infrastructure, smart contract, and application attacks[134].

5.1 Infrastructure-layer Attacks

The infrastructure-layer attacks include vulnerabilities in the BC design. They might directly target the BC, its nodes, and consensus mechanisms on which DeFi protocols operate (BC Attack) or exploit design faults in processing the transaction (MEV Attack).

5.1.1 Blockchain Attacks. The 51% attack allows tampering with the ledger data by controlling over half of the network hash rate. Other examples include block timestamp manipulation. DeFi service customers and arbitrageurs are partially affected during the BC attacks, as they are only restricted from using the DeFi Protocols. LPs lose all tokens locked on the victim BC. Similarly, issuers of stablecoins and liquids staking tokens lose the collateral. The governance users are fully affected, as their DeFi protocol is based on the victim BC. The BC attacks might be mitigated by using DeFi protocols only from the established BCs with vast networks of nodes and staked assets.

5.1.2 *Collapse of Terra Blockchain.*

The Terra blockchain belonged to the ten largest blockchains with a market capitalization of over $30bn. Its collapse was caused by the vulnerability in the economical design of its algorithmic stablecoin UST, [79], and led to the fall of the DeFi protocols operating on Terra. In infrastructure layer attacks, such as the attack on Terra, all DeFi stakeholders - service customers, liquidity providers (LPs), arbitrageurs, and governance users - are fully exposed, as the attack ultimately leads to the loss of value of the victim blockchain with all deployed smart contracts and tokens.

5.1.3 *MEV Attacks.*

MEV attacks (Miner Extractable Value[78] attacks) refers to miners or validators benefiting from their access to information about an upcoming transaction. The traditional financial market has been encountering similar cases since the 1970s[81]. The blockchain, by design, allows validators to decide on the order of transactions in the block. In MEV attacks, transactions are re-ordered for the benefit of validators at the expense of the initiator of the victim transaction. Flash loans additionally increase the arbitrage revenue of the attackers [119]. MEV Attacks affect the service customers and arbitrageurs at CLOB and AMM DEXs. Some liquid staking protocols incorporate revenue from MEV attacks in the rewards tokens[43]. Depending on the re-ordering strategy, the MEV attacks can be classified as follows:

- *Front-running attack* - the validator inserts the transaction before the victim transaction; if the victim transaction does not settle, the attack is fatal.
- *Back-running attack* - the validator's transaction is executed after the victim transaction
- *Sandwich attack* combines front- and back-running, as depicted in algorithms 5, 6.

Algorithm 5 Sandwich Price Attack[134]

1: $User_A$ sets an order to buy token i in exchange of token j with current spot price $p_{i,j}$ on an AMM with gas fee g_1.
2: $User_B$ observes the mempool and sees the transaction.
3: $User_B$ front-runs the transaction by buying x_B of token i with a higher gas fee $g_2 > g_1$ on the same AMM.
4: $User_B$ and $User_A$'s transactions are executed sequentially at respective average price of $p^B_{i,j}$ and $P^A_{i,j}$, pushing XYZ's spot price up to $p^*_{i,j}$, where $p^* > p^A > p^B > p$ due to slippage.
5: $User_B$ back-runs by selling x_B of token i at an average price of p^{B*}, with $p^* > p^{B*} > p^B$ due to slippage

DeFi protocols cannot prevent MEV attacks, as they are initiated in the blockchain infrastructure layer. Multiple approaches to mitigate transaction-ordering attacks have emerged, but all provide unsatisfactory results[92]. Considering swap fees, slippage tolerance, and gas fees, MEV attacks are only profitable if the size of the victim transaction exceeds a certain threshold[134]. Privacy-preserving blockchains with TEE[141], *e.g.*, Integritee[15] or Secret Network[20], prevent the transaction reordering from the infrastructure layer.

5.2 Smart-contract Layer-attacks

Smart contract attacks exploit vulnerabilities in DeFi protocols caused by coding mistakes, unsafe calls to untrusted contracts, and access control mistakes. The audit of the DeFi protocol leads to a decrease in the probability of exploitation by a factor of four.[139]. *Bug bounties* - reward programs for identifying software vulnerabilities - are a popular way to help to prevent smart contract exploits[62]. Smart-contract Layer attacks affect all DeFi stakeholders.

Algorithm 6 Sandwich LP Attack[134]

1: $User_A$ places a transaction order to buy x_A of token i in exchange of token j with gas fee g_1 on an AMM with reserves R_i, R_j.
2: LP_B observes the mempool and sees the transaction.
3: LP_B front-runs by withdrawing liquidity from reserves R_i and R_j with a higher gas fee $g_2 > g_1$.
4: LP_B and $User_A$'s transactions are executed sequentially, resulting in a new composition of the pool.
5: LP_B back-runs by re-providing the liquidity for R_i and R_j. Due to the transaction, the LP needs to provide less of token j for the same share in the liquidity pool.
6: LP_B sells token j for some token i as profit.

5.3 Application Layer-attacks

Application layer attacks seek to exploit to vulnerabilities in the design of the DeFi protocols and affect all DeFi stakeholders.

5.3.1 Oracle price manipulation. Oracle is a mechanism to retrieve the off-chain data, *e.g.*, ETH/USD, which can be later utilized by DeFi protocols smart contracts[107]. A centralized oracle relies on a trusted third party as a data provider. Decentralized oracles seek the information on-chain, *e.g.*, on DEXs, but are prone to price manipulation attacks 7.

Algorithm 7 Flash-loan-funded price oracle attack[134]

1: Take a flash loan to borrow x_i of token i from a lending platform, whose value is equivalent to x_j of token j at market price.
2: Swap x_i of token i for $x_j - \Delta_j$ of token j on an AMM, pushing the spot price $p_{j,i}$ down.
3: Borrow $x_i + \Delta_i$ of token i with $x_j - \Delta_j$ of token j as collateral on a lending platform that uses the AMM as their sole price oracle.
4: Repay the flash loan with x_i of token i and gain a profit of Δ_i of token i.

5.4 DeFi Risks

DeFi stakeholders 4 are exposed to various DeFi risks, *e.g.*, liquidations, de-peg, impermanent loss, and slippage risk 3 that are specific to the underlying DeFi algorithm. The risk impact differs between the stakeholders and affects in various levels service customers, LPs, arbitragers and governance users. Flash loans further amplify the losses of the victim and increase the profits of the attackers.

5.4.1 Rug pull. A rug pull is a malicious activity in which developers abandon the project and use the capital raised from the investors as a profit [110]. It affects permissionless DEXs, which do not verify the listed tokens. At the interest rate protocols, crypto-backed stablecoins require the DAO vote to register new tokens as collateral, they are resistant to rug pull attacks. Bridges and liquid staking protocols depend only on the native tokens, which can not be rug pulled.

5.4.2 Slippage risk. Slippage, the difference between the spot and realized price is a consequence of AMM, and therefore occurs only at AMM DEX. Slippage risk affects only users executing trades - service customers and arbitrageurs - and does not concern LPs. It depends on the size of AMM pool and affects all token pairs. Slippage losses can be amplified by MEV attack.

5.4.3 Impermanent Loss. Impermanent loss is a consequence of AMM. It affects only LPs to AMM DEX and services customers of yield farming protocols that employ liquidity provisions to AMM DEX in their strategy. Impermanent loss

Algorithm 8 Rug Pull Attack[134]

1: **Mint** a new token i.
2: **Create** a liquidity pool with x_i of token i and x_j of a valuable token j on an AMM.
3: **Attract** unwitting traders to buy token i with token j from the pool, effectively changing the composition of the pool.
4: **Withdraw** liquidity from the pool, and obtain $x_i - \Delta_i$ of token i and $x_j + \Delta_j$ of token j to take a profit of token j of the valuable token.

is amplified by crypto-currency volatility and can be avoided by providing liquidity to the token pairs with equal or similar values: ısynthetic tokens targeting the same value (*e.g.*, pair of USD stablecoins, SOL wrapped tokens), or *(ii)* synthetic tokens and their target tokens (*e.g.*, native tokens and rebase liquid staking tokens).

5.4.4 Liquidation risk. Liquidation risk affects service customers of DeFi lending protocols - interest rate protocols and crypto-backed stablecoin - and is amplified by crypto-currency volatility.

5.4.5 De-peg risk. refers to the circumstances when the synthetic token loses the peg to the reference value. Consequently. It affects all holders of synthetic tokens - stablecoins, wrapped tokens (bridges) and liquid staking tokens. Particularly, it concerns LPs at DEX that provide liquidity for trading synthetic assets.

6 DISCUSSION

This section starts with lessons learned from the analysis of DeFi protocols regarding the TVL measure and differentiation between CeFi and DeFi. Further, we present the current trends in DeFi protocol developments and conclude with key challenges that need to be addressed before DeFi gains wider DeFi adoption.

6.1 Lessons learned

DeFi stakeholders, especially service customers and LPs need to be aware of the counter-party risk related to the DeFi protocols. Clear differentiation between DeFi and CeFi is not straightforward nor clearly communicated by the protocols. The DeFi risks - rug pull, slippage, impermanent loss, liquidation, and de-peg - need to be considered. Moreover, the following aspects need consideration as well.

6.1.1 TVL as a measure of success. TVL, often used as a success measure in the DeFi, might be misleading for several reasons. First, TVL is a measure of market-making capital locked in DeFi, rather than user activity or traded volume. Second, some protocols, such as DEX aggregators, do not require locking any capital in order to provide services to customers (*e.g.*, optimal exchange prices at DEXs). Last, with the emergence of new algorithms *e.g.*, concentrated liquidity AMM, higher trading volumes at DEXs can be achieved with less capital locked in AMM pools.

6.1.2 CeFi vs DeFi. The framework for classification between CeFi or DeFi[118] is not always straightforward to apply. Some protocols, *e.g.*, CLOB DEX, fiat-backed stablecoins or tokenized real word assets (RWA), rely both on the blockchain security and integrity as well as on off-chain counterparties. Furthermore, some DeFi protocols, *e.g.*, crypto-backed stablecoins rely on the RWA assets and fiat-backed stablecoins, as they are included in the on-chain reserves.

6.2 Current trends

DeFi asset management concentrates on optimizing liquidity provision strategies. AMM DEX migrates to CL AMM model for higher capital efficiency. Yield farming protocols look for optimal capital allocation between AMM pools. DeFi vaults, also referred to as indexes are the new category of DeFi protocols that allow participants in the token pool, allocated by the vault manager.

6.2.1 Concentrated Liquidity AMM. After the introduction of concentrated liquidity AMM, the liquidity provisions strategy at CLAMM DEXs has attracted significantly more TVL compared to other yield farming strategies, $5.57b and $1.15b respectively [11]. Tokens in a CLAMM DEX are deployed more efficiently, compared to AMM DEX, and allow for higher transaction volume with lover TVL. Examples are the two largest DEXs on Solana: the largest DEX in terms of transaction volume is the second largest DEX in terms of TVL.

6.2.2 DeFi Asset Management. There are many emerging DeFi protocols in the area of asset management. However, their TVL is still low. Two categories worth mentioning included *indexes* and *reserve currencies*. The indexes protocols, such as SetProtocol[44], Enzyme[31], allow DeFi users to create, manage and allocate tokens to vaults, which are similar in their function to the funds and ETF in TradFi. Depositors transfer liquidity into the vault, whereas vault managers allocated tokens to various DeFi protocols to generate yield. Reserve currency protocols function similarly. Depositors transfer tokens into the managed vault. The difference is the valuation of the vault. In the case of the index protocols, the value of the vault is equal to the value of all tokens in the vault. In the case of the reserve currency vault, the value is determined by supply and demand.

6.3 Future Challenges

As presented in the risk sections, there are many open challenges hindering the wider DeFi adoption. Impermanent loss makes the liquidity providing for new tokens unattractive and consequently increase the cost of new token launch. The major obstacle to the lending protocol is over-collateralization which leads to lower capital efficiency and liquidation risk of the collateral.

6.3.1 Impermanent Loss at AMM DEX. AMM-based DEX differentiates by utilizing various reserve curves and swap fees, pool weightage, and liquidity mining, mainly in order to mitigate the impermanent loss for the LPs. Liquidity mining refers to the process of collecting native tokens of the DeFi protocol in exchange for participating in the liquidity pool and is a form of subsidizing AMM pools for the token designers. Most DEXs, like Uniswap[51] or Balancer[109], offer 50/50 pool weightage, in which LPs must supply the equal value of each token in the liquidity pool. Whereas Balancer[109] offers a variable pool supply criteria. ON Curve[80], the pool weightage is dynamic and does not rebalance its pool. Pool weightage impacts impermanent loss. Further research on mitigation of impermanent loss, by applying other AMM formula, or using on-chain derives, is required.

6.3.2 Over-collateralized Lending. Although DeFi over-collateralized lending mitigates the risks arising from the anonymous borrowers, it exposes the borrower to the liquidation risk and does not allow to use of the digital asset locked as collateral. Decentralized prime brokerage and on-chain identity - present alternative models for the DeFi undercollateralized lending [60]. *Decentralized prime brokerage* protocols - Oxygen[40], DeltaPrime[29], Gearbox[32] - retain the control over the use of the borrowed funds and identity-based protocols - TrueFi[48], Goldfinch[34] provide the mapping between on-chain borrowers and real-world entities.

6.3.3 Security vs Credit. Lending protocols compete with the staking process in PoS BC, consequently damaging the network security [69, 72, 73]. When the yield provided by the lending protocols is more attractive than the staking rate (inflation rate), stakers are incentivized to remove their staked tokens and lend them out, thus reducing the network security [72]. MEV-redistribution improves the network security by redistributing MEV revenue and disincentivizing to unstake[73]. Some liquid staking protocols, *e.g.*, RocketPool[43], include in their tokens MEV rewards, generated by the validators. The risk of liquid staking includes not only slashing risk, typically for staking, but also de-peg risk, inherent to the synthetic tokens.

6.3.4 Bridges. Full collateralization of wrapped tokens is inefficient, but over-collateralization is required due to the volatility of digital assets [63]. With progress in blockchain interoperability and scalability, multi-chain DEX emerged such as THORChain[1]. They offer swaps of digital assets between BCs.

6.3.5 Synthetic assets de-peg. In 2023, the liquid staking protocol Lido[3] overtook the decentralized stablecoin protocol MakerDAO [108], and became the largest DeFi protocol in terms of TVL[11]. As indicated in this work, little research has been performed on synthetic assets other than stablecoins. Wrapped tokens and liquid staking tokens are gaining popularity in DeFi, but are exposed to de-peg risk amond other. The stability of their promised peg, especially the inclusion of staking rewards, requires empirical examinations.

7 SUMMARY

This paper analyzes the major categories of DeFi protocols that represent over 89% of TVL. Both the technical and economical design of DeFi protocols are examined. Our analysis shows that any DeFi protocol can be classified into one of three classes: *(i)* liquidity pool protocols *(ii)* synthetic token protocols or *(iii)* aggregator. Liquidity pool protocols manage the supply and demand sites with smart contracts, *e.g.*, interest rate protocols manage supply and demand for loanable funds, and vAMM / AMM DEXs manage supply and demand for tradeable tokens and derivatives. Synthetic token protocol issues tokens with a value pegged to the reference value, *e.g.*, stablecoins (peg to the fiat currency), wrapped tokens (peg to the tokens on other blockchains), liquid staking tokens (peg to the staked tokens, and staking rewards), or index tokens. Aggregators optimize the liquidity provision strategies across DeFi protocols (yield farming protocols) or best trade execution at DEX (DEX aggregators). Our taxonomy for the chain architecture of DeFi protocols suggests that DeFi cannot be defined as a smart contract-enabled financial application, but a more general definition is required that is based on the security and integrity of blockchains. Finally, we showed that the risk in DeFi depends on three dimensions: *(i)* DeFi agents, *(ii)* DeFi protocols, and *(iii)* tokens involved. We indicated specifically that service customers and LPs are exposed to different risks, and aggregator protocols accumulate the risk of the protocols and tokens they utilize.

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